Local stability of a two-copy entanglement-of-purification ansatz at the von Neumann boundary

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

Entanglement of purification requires an optimization over purifications, and its additivity at von Neumann order remains open even for classical states. For a diagonal two-qubit family recently studied at other Rényi orders, we analyze two copies at $r = 1/2$ in the minimal 4×4 ancillary split. We prove, with a computer-assisted interval certificate, that the product-canonical purification is a strict local minimum modulo local ancillary unitaries throughout the full complex $U(16)$ flag manifold, including rank-opening directions. The proof combines an 81-dimensional quotient-Hessian bound with a logarithmic tail estimate. No global optimality, enlarged-ancilla, or additivity claim is made.

Keywords: entanglement of purification; von Neumann entropy; purification landscape; interval arithmetic; local minimum

1 Introduction

The entanglement of purification (EoP) of a bipartite state ρ_{AB} is the least entanglement entropy across $AA' : BB'$ among all of its purifications [1]. Its regularization has operational meanings in correlation preparation and visible compression [1, 2], but it is not known in general whether the one-copy quantity is additive. Numerical work on two-qubit Werner states supplied evidence for nonadditivity [3], without resolving the von Neumann problem analytically.

Negari and Baghali Khanian recently isolated a simple diagonal two-qubit family for which the one-copy optimization can be solved exactly at every Rényi order [4]. For two copies they proved strict nonadditivity for every order below one using a signed-canonical witness, and proved strong additivity, under their stated classical premise, for orders at least two. The interval $1 \leq \alpha < 2$, including the von Neumann point $\alpha = 1$, remains open in that work. Their finite sign search does not cover continuous complex purifying flags, while a positive ordinary Hessian alone cannot settle directions that open zero eigenvalues of the reduced state.

Here we resolve one deliberately local part of that open landscape. At the exact state parameter $r = 1/2$, and only in the minimal ancillary split $(d_{A'}, d_{B'}) = (4, 4)$, we show that the product-canonical two-copy purification is a strict local minimum modulo local ancillary unitaries over the complete complex flag manifold $U(16)$. The tangent space separates into 112 rank-preserving and 144 support-opening directions. Removing the 31-dimensional ancillary gauge orbit leaves an 81-dimensional analytic Hessian. An outward-rounded ball-arithmetic calculation certifies its least eigenvalue to exceed 0.159. A separate Schur-complement argument uses the $-t^2 \log t^2$ entropy generated at the rank boundary to make this Hessian result uniform for mixed rank-preserving and support-opening paths.

The conclusion is local and dimension-specific. It neither minimizes over distant points of $U(16)$ nor compares larger ancillary splits, and therefore does not prove two-copy additivity. Its role is instead to identify a rigorous obstruction: within the minimal split, a von Neumann improvement at this state cannot bifurcate infinitesimally from the canonical purification.

2 Model and main result

For $0 \le r \le 1$, consider the classical state

$$\rho_r = \frac{r^2|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |11\rangle\langle 11|}{3 + r^2}. \tag{1}$$

All logarithms below have base two, so entropies are measured in bits. The von Neumann EoP is

$$E_P(\rho_{AB}) = \inf_{|\Psi\rangle:\, \mathrm{Tr}_{A'B'}|\Psi\rangle\langle\Psi| = \rho_{AB}} S(\mathrm{Tr}_{BB'}|\Psi\rangle\langle\Psi|)\,, \qquad S(\sigma) = -\mathrm{Tr}(\sigma \log_2 \sigma). \tag{2}$$

For two copies, write $I = (i_1, i_2)$ and $J = (j_1, j_2)$, each ordered lexicographically as $00, 01, 10, 11$, and let $P_{IJ} = p_{i_1 j_1} p_{i_2 j_2}$. At $r > 0$ all sixteen probabilities are positive. In the minimal split $\mathcal{H}_{A'} \otimes \mathcal{H}_{B'} \cong \mathbb{C}^4 \otimes \mathbb{C}^4$, every orthonormal flag family is an orthonormal basis and hence can be represented by $W \in U(16)$:

$$|\Psi_W\rangle = \sum_{I,J} \sqrt{P_{IJ}}\, |I\rangle_A |J\rangle_B W |IJ\rangle_{A'B'}. \tag{3}$$

Let

$$F(W) = S(\mathrm{Tr}_{BB'}|\Psi_W\rangle\langle\Psi_W|)\,. \tag{4}$$

The product-canonical purification is $W_0 = I_{16}$.

At $r = 1/2$, the normalized one-copy coefficient matrix across $AA' : BB'$ is

$$C_1 = \frac{1}{\sqrt{13}} \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}, \qquad C = C_1 \otimes C_1. \tag{5}$$

The eigenvalues of $C_1 C_1^\dagger$ are

$$\lambda_\pm = \frac{13 \pm 3\sqrt{17}}{26}. \tag{6}$$

Consequently the four nonzero Schmidt weights at W_0 are $\lambda_-^2, \lambda_-\lambda_+, \lambda_-\lambda_+, \lambda_+^2$, and

$$F(W_0) = 2h_2(\lambda_+) = 0.329435221224950\ldots. \tag{7}$$

The one-copy optimality needed to identify this as twice the one-copy EoP is the exact result of Ref. [4]; the local theorem below does not reprove that global one-copy statement.

Local unitaries on the ancillary factors preserve the entropy. Define their image and orbit by

$$\mathcal{G} = \{U \otimes V : U, V \in U(4)\}, \qquad \mathcal{O} = \mathcal{G} W_0. \tag{8}$$

Theorem 1 (minimal-split local stability)**.** *For the state $\rho_{1/2}$ in Eq. (1), there is a neighborhood $\mathcal{N}$ of $\mathcal{O}$ in $U(16)$ such that*

$$F(W) \ge F(W_0) \quad (W \in \mathcal{N}), \tag{9}$$

with equality only for $W \in \mathcal{O}$ after $\mathcal{N}$ is reduced if necessary. Thus W_0 is a strict local minimum modulo $\mathcal{G}$ among all complex orthonormal flags in the fixed $(4, 4)$ ancillary split, including paths that open new Schmidt support.

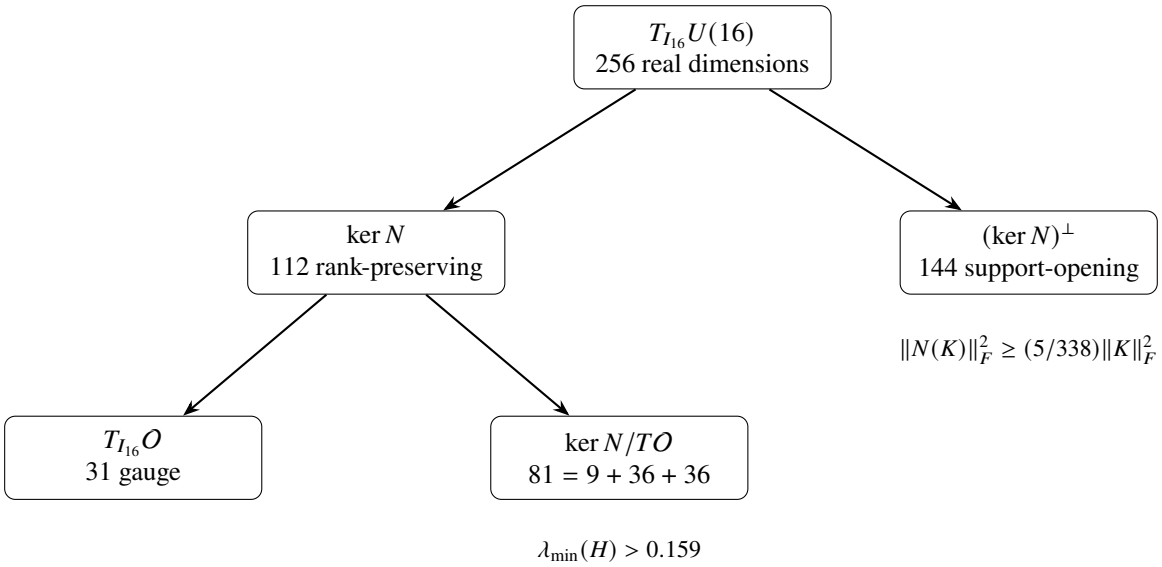


Figure 1: Exact tangent decomposition at the product-canonical point. The ordinary Hessian acts on the 81-dimensional fixed-rank quotient; the 144-dimensional complement is controlled through the support-opening map and the logarithmic entropy tail.

3 Tangent geometry and gauge quotient

Index a flag input by $x = (I, J)$ and a flag output by $y = (u, v)$, with all four indices in $\{0, 1, 2, 3\}$. Across $AA' : BB'$, the 16×16 coefficient matrix of Eq. (3) is

$$M(W)_{(I,u),(J,v)} = C_{IJ} W_{(u,v),(I,J)}, \qquad F(W) = S(MM^{\dagger}). \tag{10}$$

At W_0, only the rows (I, I) and columns (J, J) are active; their 4×4 block is the invertible matrix C.

Let $K^{\dagger} = -K$ be a tangent and use $W(t) = \exp(tK)$. If P and Q project onto the active row and column spaces, the normal–normal part of the first variation is the real-linear map

$$N(K)_{(I,u),(J,v)} = C_{IJ} K_{(u,v),(I,J)}, \qquad u \neq I, \quad v \neq J. \tag{11}$$

Because every C_{IJ} is nonzero, $N(K) = 0$ precisely when no off-diagonal entry of K connects two points of the 4×4 flag grid whose two coordinates both change.

This gives an exact count. The kernel contains 16 imaginary diagonal generators. There are 24 unordered grid edges sharing I and 24 sharing J, each with real and imaginary anti-Hermitian generators. Hence

$$\dim_{\mathbb{R}} \ker N = 16 + 2(24 + 24) = 112, \qquad \dim_{\mathbb{R}} (\ker N)^{\perp} = 144. \tag{12}$$

The tangent to O consists of $X \otimes I_4 + I_4 \otimes Y$ with $X, Y \in \mathfrak{u}(4)$. The kernel of $(X, Y) \mapsto X \otimes I + I \otimes Y$ is $(\mathrm{i}cI, -\mathrm{i}cI)$, so this gauge tangent has dimension 31 and lies in $\ker N$.

For completeness, an exact orthonormal complement of the gauge tangent in $\ker N$ is constructed from the three Helmert contrasts

$$h_1 = \frac{(1, -1, 0, 0)}{\sqrt{2}}, \quad h_2 = \frac{(1, 1, -2, 0)}{\sqrt{6}}, \quad h_3 = \frac{(1, 1, 1, -3)}{\sqrt{12}}. \tag{13}$$

It contains nine diagonal phase tables $h_p h_q^{\mathsf{T}}$, 36 same-I rook-edge contrasts, and 36 same-J rook-edge contrasts. Thus the physical fixed-rank quotient has dimension $9 + 36 + 36 = 81$. Figure 1 summarizes the decomposition.

The 144 normalized opening generators have mutually orthogonal images under N. For an unordered edge between $x = (I, J)$ and $y = (u, v)$ with both coordinates different, either its real or imaginary normalized generator has

$$\|N(K)\|_F^2 = \frac{C_{IJ}^2 + C_{uv}^2}{2}. \tag{14}$$

Exact enumeration gives

$$\min_{K \in (\ker N)^\perp,\, \|K\|_F = 1} \|N(K)\|_F^2 = \frac{5}{338}. \tag{15}$$

In particular, the opening map is quantitatively injective on its 144-dimensional complement.

The point W_0 is stationary in every tangent direction. For a diagonal $K_{(I,J),(I,J)} = \mathrm{i}\theta_{IJ}$, the active part of $\rho'(0)$, where $\rho = MM^\dagger$, is purely imaginary and antisymmetric after removal of the factor i, whereas $\log \rho(0)$ is real symmetric. Their trace pairing vanishes. Rook-edge directions have only active–normal first variations, and the remaining directions have normal–normal first variations, so they also have zero first entropy variation. In an opening direction the new weights are $O(t^2)$ and their entropy is $O(-t^2 \log t^2)$, whose first derivative at zero is again zero.

4 Certified Hessian on the fixed-rank quotient

We equip $\mathfrak{u}(16)$ with $\langle K, L\rangle = \mathrm{Re}\,\mathrm{Tr}(K^\dagger L)$. Write

$$M(t) = M_0 + tM_1 + \frac{t^2}{2} M_2 + O(t^3), \quad M_1 = L(K), \quad M_2 = L(K^2), \tag{16}$$

where L is the linear coefficient map in Eq. (10). Then

$$\rho_1 = M_1 M_0^\dagger + M_0 M_1^\dagger, \tag{17}$$

$$\rho_2 = M_2 M_0^\dagger + M_0 M_2^\dagger + 2 M_1 M_1^\dagger. \tag{18}$$

Let P now denote the four-dimensional positive support of ρ_0 and $Q_0 = I - P$. Let U diagonalize $P\rho_0 P$, with eigenvalues $\{\lambda_a\}_{a=1}^4 = \{\lambda_-^2, \lambda_-\lambda_+, \lambda_-\lambda_+, \lambda_+^2\}$. Put

$$X = U^\dagger P \rho_1 P U, \quad Z = U^\dagger P \rho_2 P U, \quad Y = Q_0 \rho_1 P U. \tag{19}$$

For $f(x) = -x \log_2 x$, define the divided difference

$$f'^{[1]}(x, y) = \begin{cases} [f'(x) - f'(y)]/(x - y), & x \neq y, \\ f''(x), & x = y. \end{cases} \tag{20}$$

Hermitian spectral perturbation, including the double eigenvalue $\lambda_-\lambda_+$, gives the second directional derivative

$$q(K) = \sum_a f'(\lambda_a) Z_{aa} + \sum_{a,b} f'^{[1]}(\lambda_a, \lambda_b) |X_{ab}|^2 + 2 \sum_a \frac{f'(\lambda_a)}{\lambda_a} \|Y_{:a}\|_2^2. \tag{21}$$

The last term records motion of the positive eigenspace into the zero eigenspace. If $K \in \ker N$, the small singular values of $M(\exp(tK))$ are $O(t^2)$, so their squared Schmidt weights are $O(t^4)$ and their entropy is $O(-t^4 \log |t|)$. They therefore have zero second derivative. Thus Eq. (21) is the Hessian of the full entropy on $\ker N$.

The 81-by-81 matrix H of q in the basis described after Eq. (13) was evaluated with 320-bit Arb midpoint-radius balls through `python-flint`; Arb implements outward-rounded ball arithmetic [5].

Table 1: Interval-certificate summary. All inequalities in the certificate use outward-rounded balls; the midpoint eigensolver is not part of the proof.

Quantity	Certified or recorded value
Quotient dimension	$81 = 9 + 36 + 36$
Arithmetic precision	320 bits
Largest Hessian-entry radius	$< 1.64 \times 10^{-93}$
Rational shift	$159/1000$
Strictly positive LDL^{T} pivots	81 of 81
Smallest pivot lower endpoint	> 0.0290093
Reconstruction residual	every entry encloses zero
Opening-map least squared scale	$5/338$ exactly
Midpoint $\lambda_{\min}$ (diagnostic)	0.159148973002032
Independent precision replays	256 and 384 bits, both pass

The calculation uses exact rational and radical input expressions and analytic divided differences, not finite differences. It then performs an interval, no-pivot LDL^{T} factorization of

$$H - \frac{159}{1000} I_{81}. \tag{22}$$

Every interval pivot is strictly positive. Sylvester's criterion therefore certifies

$$\boxed{\lambda_{\min}(H) > 0.159} \quad \text{bits per squared unit tangent.} \tag{23}$$

Table 1 records the decisive checks. The smallest pivot is not itself an eigenvalue bound; it certifies the factorization of the already shifted matrix. Likewise, the binary64 midpoint eigenvalue is a diagnostic only.

5 Uniform control of support opening

Positive definiteness on $\ker N$ does not by itself prove Theorem 1: the reduced state at W_0 has twelve zero eigenvalues. We now keep their logarithmic entropy explicitly.

Use the active and normal row and column decompositions to write, near W_0,

$$M(W) = \begin{pmatrix} A & B \\ E & D \end{pmatrix}, \qquad A(W_0) = C, \tag{24}$$

and define the analytic Schur complement

$$\mathcal{Z}(W) = D - EA^{-1}B. \tag{25}$$

Its derivative at W_0 is N. The exact factorization

$$M = \begin{pmatrix} I & 0 \\ EA^{-1} & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & \mathcal{Z} \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & I \end{pmatrix} \tag{26}$$

has uniformly bounded outer factors and inverses in a sufficiently small neighborhood. By the characterization of the Frobenius distance to matrices of rank at most four, the total small-weight mass

$$\tau(W) = \sum_{j=5}^{16} s_j(M(W))^2 \tag{27}$$

satisfies $\tau(W) \geq c_0 \|\mathcal{Z}(W)\|_F^2$ there for some $c_0 > 0$.

Let Π be the orthogonal projection onto the real 144-dimensional image of N and set $z = \Pi\mathcal{Z}(W)$. Work on a local slice perpendicular to the 31 gauge directions. The exact gap in Eq. (15) makes the derivative from opening coordinates to z invertible. The inverse-function theorem therefore gives local coordinates (k, z) on the slice, where $k \in \mathbb{R}^{81}$ is the fixed-rank quotient coordinate, and after shrinking the chart,

$$\tau(k, z) \geq c\|z\|^2. \tag{28}$$

Let $G(k, z)$ be the entropy contribution of the four eigenvalues separated from zero. This is analytic despite their internal double eigenvalue. Full stationarity and Eq. (23), followed by an ordinary Taylor bound on the remaining Hessian blocks, yield constants $A, B > 0$ such that

$$G(k, z) - G(0, 0) \geq A\|k\|^2 - B\|k\|\|z\| - B\|z\|^2. \tag{29}$$

For sufficiently small τ, the entropy T of the other twelve weights obeys

$$T = \sum_{j=5}^{16} -s_j^2 \log_2 s_j^2 \geq -\tau \log_2 \tau \geq -c\|z\|^2 \log_2(c\|z\|^2). \tag{30}$$

The first inequality uses $s_j^2 \leq \tau$; the second uses Eq. (28) and monotonicity of $-x \log_2 x$ near zero.

Young's inequality absorbs the mixed term in Eq. (29) into half of the positive k term plus a constant multiple of $\|z\|^2$. The logarithm in Eq. (30) uniformly dominates every such bounded quadratic penalty as $z \to 0$. Hence $F(k, z) > F(0, 0)$ at every sufficiently small nonzero point of the slice. Equality occurs only at its origin. Finally, the compact ancillary action admits a tubular local slice, and F is constant on each ancillary orbit. This transfers the strict slice minimum to a neighborhood of O and completes the proof of Theorem 1.

6 Interpretation and limitations

The result controls all complex flags in the smallest bipartite ancillary space capable of holding the sixteen occupied two-copy branches. It is stronger than a scan of sign patterns or a direction-by-direction numerical test: the fixed-rank part is certified on the complete gauge quotient, and the rank boundary is controlled uniformly rather than inferred from an ordinary Hessian.

At the same time, the statement is intentionally narrow. The neighborhood radius is existential and has not been optimized. Only $r = 1/2$ and $\alpha = 1$ are treated. A purification at finite distance in $U(16)$ could still have smaller entropy, and a larger ancillary split has additional directions absent here. Thus neither $E_P(\rho_{1/2}^{\otimes 2}) = 2E_P(\rho_{1/2})$ nor any regularized correlation cost follows. The rigorous consequence is instead a landscape constraint: at this state and in this split, any von Neumann improvement must occupy a separate finite-distance basin or enter through ancillary enlargement.

Two natural next steps are to turn the existence proof into an explicit certified neighborhood radius, and to repeat the tangent-plus-boundary analysis in a padded ancillary embedding. Either extension requires new scientific certification and is not assumed here.

7 Reproducibility and data availability

The accompanying source package contains the exact basis construction, the analytic Hessian evaluator, the 320-bit Arb certificate, its 256- and 384-bit precision replays, independent finite-difference diagnostics, and the source baseline reconstruction. The designated certificate is generated from the project root by

```
python3 evidence/certify_u16_hessian.py
```

and must report 81 positive pivots for the shift in Eq. (22). The recorded environment uses Python 3.12.14, `python-flint` 0.9.0, NumPy 2.3.5, SciPy 1.18.1, and mpmath 1.3.0. The reproducibility README distinguishes the rigorous interval output from diagnostic floating-point calculations. No experimental or personal data were used. A permanent public repository and identifier remain decisions for the author before submission.

Declarations

AI use. Generative AI (OpenAI Codex, GPT-5.6 Sol, accessed September 2026) assisted with formulating the scoped question, mathematical derivations, code, literature checks, computational verification, an independent AI referee replay, and manuscript drafting. AI is not an author and cannot assume accountability. The named author must independently review the claims, proof, code, citations, and prose, and approve an accurate final declaration before any submission.

Author contributions, funding, and competing interests. These declarations require confirmation and completion by the author before submission. No external submission has been made from this draft.

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